Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/25/2005
PI Name:	Wayner Jr., Peter Ph.D.		
Project Title:	Theoretical and Experimental Investigation of the Stability of an Evaporating Constrained Vapor Bubble		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	wayner@rpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	518-276-6199
Organization Name:	Rensselaer Polytechnic Institute		
PI Address 1:	Isermann Department of Chemical Engineering		
PI Address 2:	110 8th Street		
PI Web Page:			
City:	Troy	**State:**	NY
Zip Code:	12180	**Congressional District:**	21
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	04/15/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Wayner, Jr., Peter C.		
COI Name (Institution):	Plawsky, Joel (Rensselaer Polytechnic Institute) Harris, Wong (Louisianna State University)		
Grant/Contract No.:	NAG3-2351		
Performance Goal No.:	2.6		
Performance Goal Text:	Advance the state of knowledge sufficiently to allow development of reliable and efficient heat transfer technology for space and extraterrestrial operations.		

Task Description:	A transparent quartz Constrained Vapor Bubble Heat Exchanger (CVBHX) was used in the earths environment to obtain experimental data on stability and heat transfer for both completely wetting (pentane) and partially wetting (ethanol) fluids. The details of the flow field, as a function of heat flux, were obtained through a microscope using an Image Analyzing Interferometer (IAI) to obtain the pressure field. Using a two dimensional model of an evaporating meniscus, a complementary linear stability analysis of the evaporating meniscus is being done by perturbing the interfacial position. Professor Wong's group at Louisiana State University is doing the stability analysis and Professors Plawsky and Wayner's group at Rensselaer is doing the experimental portion. The objective of this effort is to better understand the stability and physics of evaporation and condensation as they affect the heat transfer processes in a constrained vapor bubble heat exchanger (CVBHX). This CVBHX consists of a small-enclosed container partially filled with a liquid. When a temperature difference is applied to the ends of the CVBHX, evaporation occurs at the hot end and condensation at the cold end with fluid circulation due to interfacial forces. This results in a very effective heat transfer device with great potential in space applications. As the applied heat flux increases, the evaporating meniscus is first stable and then starts to oscillates and then becomes unstable. This grant is a complementary extension of NAG3-2383. This ground-based study complements the planned experimental study of the CVBHX on the International Space Station (NAG3-2383). It is anticipated that the contact line region of the CVBHX will become unstable at high heat fluxes during the flight experiments. Therefore, it is prudent to develop the ability to analyze these anticipated unique data on instability using well-developed techniques for the analysis of stability, which are the objectives of this task.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The contract has been completed and the final report has been submitted.
Bibliography Type:	Description: (Last Updated: 09/25/2012)
Articles in Peer-reviewed Journals	Zheng L, Wang YX, Wayner PC Jr, Plawsky JL. "Microscale transport processes in the evaporator of a constrained vapor bubble." Journal of Thermophysics and Heat Transfer. 2003 Apr-Jun;17(2):166-73. , Jun-2003
Articles in Peer-reviewed Journals	Plawsky JL, Panchamgam SS, Gokhale SJ, Wayner PC Jr, DasGupta S. "A study of the oscillating corner meniscus in a vertical constrained vapor bubble system." Superlattices and Microstructures. 2004 Mar-Jun;35(3-6):559-72. http://dx.doi.org/10.1016/j.spmi.2003.11.010 , Jun-2004
Articles in Peer-reviewed Journals	Zheng L, Plawsky JL, Wayner PC Jr, DasGupta S. "Stability and oscillations in an evaporating corner meniscus." ASME Journal of Heat Transfer. 2004 Apr;126(2):169-78. http://dx.doi.org/10.1115/1.1652046 , Apr-2004
Articles in Peer-reviewed Journals	Wu Q, Wong H. "A slope-dependent disjoining pressure for non-zero contact angles." Journal of Fluid Mechanics. 2004 May;506:157-85. http://dx.doi.org/10.1017/S0022112004008420 , May-2004
Presentation	Wu, Q. and H. Wong "A slope-dependent disjoining pressure valid for non-zero contact angles" MRS Fall Meeting, Boston, MA paper P7.9 Nov-2003
Presentation	Wu, Q. and H. Wong "A slope-dependent disjoining pressure valid for non-zero contact angles" Annual Meeting of the APS Division of Fluid Dynamics, paper MN3, East Rutherford, NJ Bulletin of the American Physical Society 48, pp. 243, , Nov-2003
Presentation	Zhang, J. and H. Wong "Fluid flow and heat transfer in an idealized micro heat pipe" Annual Meeting of the APS Division of Fluid Dynamics, paper MD4, East Rutherford, NJ Nov-2003